

CHANGDE, China

WMO: 576620

Lat: 29.1168N

Lon: 111.6770E

Elev: 151

StdP: 99.52

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	-0.1	-9.5	1.7	4.5	-6.8	2.2	4.9	6.4	4.7	5.5	5.3	1.7	0	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	36.5	26.5	35.2	26.5	34.0	26.2	28.3	33.1	27.7	32.6	27.2	31.9	2.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.3	31.2	26.5	22.4	30.6	25.9	21.7	30.1	92.7	33.3	89.9	32.7	87.2	32.0	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.5	5.4	4.5	DB	-2.5	38.1	1.6	1.1	-3.7	38.9	-4.6	39.5	-5.5	40.2	-6.6	41.0	
			WB	-3.6	29.2	1.3	0.7	-4.6	29.8	-5.4	30.2	-6.1	30.6	-7.1	31.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	5.4	7.7	12.5	18.2	22.8	26.0	29.3	28.6	24.5	19.1	13.3	7.8
	DBStd	8.88	3.38	4.21	4.44	4.26	3.33	3.01	2.95	3.08	3.48	3.50	3.89	3.15
	HDD10.0	353	147	86	24	1	0	0	0	0	0	0	14	82
	HDD18.3	1459	401	297	188	53	4	0	0	2	33	156	326	
	CDD10.0	3271	4	22	102	247	397	481	599	578	434	282	113	13
	CDD18.3	1338	0	1	7	49	143	231	341	319	186	57	5	0
	CDH23.3	13603	0	4	46	324	1037	2192	4330	3858	1537	258	17	0
	CDH26.7	6018	0	1	11	92	346	856	2228	1875	559	49	2	0
Wind	WSAvg	2.0	1.7	1.8	2.1	2.1	2.1	2.0	2.1	2.2	2.1	1.9	1.8	1.7
Precipitation	PrecAvg	1376	58	75	104	164	186	205	195	115	77	84	78	36
	PrecMax	2049	137	172	262	401	363	409	424	292	342	165	183	90
	PrecMin	778	17	7	41	47	88	65	33	16	8	14	5	1
	PrecStd	246	31	45	49	77	66	88	95	64	67	46	45	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	22.4	28.0	31.9	33.9	35.8	37.8	38.0	35.9	30.4	25.5	18.8
		MCWB	10.9	14.4	19.2	21.6	22.8	26.0	26.9	26.1	26.1	21.5	17.0	11.8
	2%	DB	14.1	18.4	24.0	29.4	32.0	34.2	36.6	36.5	33.2	27.8	22.7	16.1
		MCWB	8.9	12.4	16.3	20.9	22.8	25.8	26.7	26.5	25.3	20.1	16.0	10.2
	5%	DB	12.0	15.7	21.4	27.1	30.3	32.6	35.6	35.0	31.4	26.1	20.6	14.1
		MCWB	7.9	10.7	15.0	19.7	22.4	25.4	26.7	26.9	24.6	18.8	15.1	9.3
	10%	DB	10.3	13.7	19.1	24.9	28.5	31.1	34.3	33.5	29.8	24.4	18.8	12.5
		MCWB	7.0	9.7	14.2	18.8	21.7	24.9	26.5	26.6	23.6	18.5	14.3	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	16.1	21.0	24.3	26.0	27.9	29.0	29.1	27.4	23.9	18.9	13.1
		MCDB	15.0	21.3	26.3	29.4	30.9	32.5	34.1	33.4	32.4	27.3	22.3	16.1
	2%	WB	10.0	13.2	18.1	22.3	24.8	27.0	28.3	28.4	26.6	21.6	17.2	11.4
		MCDB	13.1	16.4	21.7	27.0	29.0	31.4	33.4	33.0	31.2	25.5	20.8	14.8
	5%	WB	8.6	11.5	16.3	20.9	23.8	26.3	27.8	27.8	25.8	20.5	16.1	10.1
		MCDB	11.4	14.7	20.1	24.9	27.8	30.8	32.8	32.5	30.3	24.3	19.3	13.3
	10%	WB	7.2	10.2	14.8	19.5	22.9	25.7	27.2	27.2	24.8	19.5	14.9	9.0
		MCDB	9.9	13.3	18.1	23.8	26.7	29.9	32.2	31.9	28.9	23.0	17.9	11.9
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	6.5	7.0	7.0	6.4	6.6	6.6	6.6	6.4	6.1	5.7
		MCDBR	9.1	10.2	10.9	10.9	10.3	8.7	8.4	8.5	8.8	9.3	9.4	9.7
		MCWBR	5.2	5.4	5.3	4.7	3.8	3.0	2.5	2.6	3.1	3.4	4.1	5.0
	5% WB	MCDBR	8.3	8.7	9.5	9.5	8.0	7.5	7.6	7.6	7.8	7.7	7.6	8.3
		MCWBR	5.1	5.0	5.3	5.1	3.9	3.2	2.9	3.0	3.6	3.7	3.8	4.7
Clear-Sky Solar Irradiance	taub	0.697	0.744	0.767	0.752	0.696	0.696	0.616	0.686	0.728	0.750	0.714	0.685	
	taud	1.465	1.422	1.414	1.483	1.621	1.645	1.842	1.661	1.554	1.463	1.494	1.506	
	Ebn at Noon	529	556	586	621	661	658	712	657	605	544	514	511	
	Edh at Noon	260	293	313	300	262	255	209	248	268	277	250	239	
All-Sky Solar Radiation	RadAvg	1.74	2.03	2.90	3.53	3.86	4.08	4.99	4.45	3.60	2.77	2.26	1.84	
	RadStd	0.31	0.53	0.26	0.43	0.55	0.53	0.52	0.41	0.41	0.42	0.37	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-1.90	N/A	-0.42	-0.55	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page